

COMMONWEALTH OF PENNSYLVANIA — PROCUREMENT ANALYSIS

Indirect Costs of a Complex Procurement System: A Shallow Vendor Pool, Reduced Competition, and the Cost to the Commonwealth

How Competitive Are PA State Contracts? · PA eMarketplace Bid Tabulation Review — FY 2024–2025

Produced by U.S. Digital Response

100 contracts manually verified · 825 total solicitations · ±10% margin of error at 95% confidence^[1] · March 2026

EXECUTIVE SUMMARY

A manual review of **100 Pennsylvania state contracts** — a statistically representative sample of 825 competitive solicitations — reveals that **1 in 4 contracts attracted only a single bidder**, and nearly half attracted two or fewer. By comparing the highest and lowest bids within each sampled PA contract, we can see that prices are **16% lower when a second bidder enters** and **68% lower when competition reaches five or more bidders**^[2]. International research across 3.5 million contracts ([Open Contracting Partnership](#)) finds that single-bid contracts are on average **7% more expensive**^[3] than contested ones — Pennsylvania contracts are twice that (16%).

1 in 4

CONTRACTS RECEIVE ONLY ONE BID

No competitive pressure on price

42%

HAVE 1–2 BIDDERS TOTAL

Formally open, under-competed

68%

PRICE REDUCTION AT 5+ BIDDERS

vs. single-bidder price ceiling

up to
~\$34M^[4]

ESTIMATED ANNUAL COST OF LOW COMPETITION

with 5+ bid competition — indicative

SECTION 1 — COMPETITION LEVELS

How Competitive Are PA Contracts?

All 100 contracts were formally tendered through the open bid process.

BIDDER COUNT DISTRIBUTION (100 CONTRACTS)



42.0%

Had 1–2 bidders

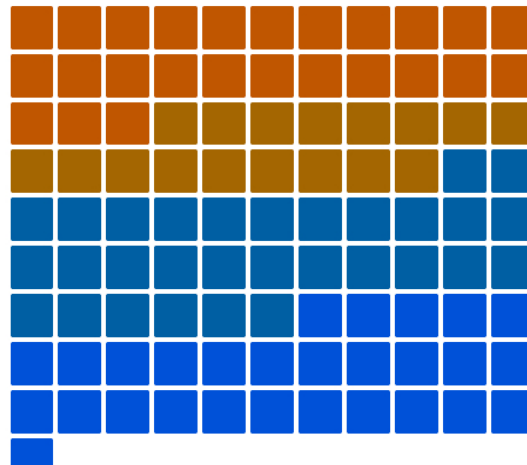
EACH SQUARE = 1 CONTRACT (100 TOTAL)

42

42.0% — 1–2 bidders under-competed

58

58.0% — 3+ bidders



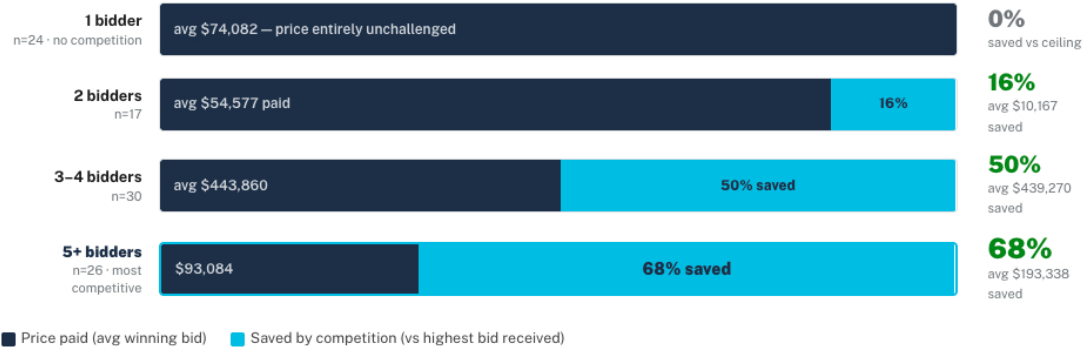


Nearly half of all formally tendered contracts — **42 of 100** in this sample — attracted only 1 or 2 bidders. These contracts went through the full procurement process but produced **little or no effective price competition**.

SECTION 2 — PRICE IMPACT OF COMPETITION

What Competition Actually Delivers

Each bar shows what Pennsylvania *paid* (dark) versus what competition *saved* (teal) — the gap between highest and lowest submitted bids. These are real PA contract outcomes, not projections.



Across this sample, the 24 single-bid contracts represent \$1.78M in entirely unchecked spending — with an estimated **\$285,000** in overpayment embedded within that figure, based on Pennsylvania's own 16% price premium on low-competition contracts. That overpayment grows substantially as competition deepens — PA data shows prices are **68% lower** when five or more bidders compete, meaning an overpayment of approximately **\$1.2M** on this sample alone.

CONSERVATIVE ESTIMATE

If 1-2 bidder contracts reached only 3-4 bidder pricing, the estimated annual saving is approximately:

~\$10M

per year — 3-4 bid competition
still below the fully competitive benchmark

ANNUAL SAVINGS IMPLICATION

If under-competitive contracts (1-4 bidders) achieved 5+ bidder pricing across the full 825-contract portfolio, estimated annual saving across PA procurement is approximately:

~\$34M

per year — indicative estimate

SECTION 3 — AGENCY PERFORMANCE

The Performance Gap

Every agency operates under the same procurement rules. The variation in outcomes reflects outreach and process — not market conditions beyond any agency's control.

AGENCY	CONTRACTS	SINGLE BID %	1-2 BID %	AVG BIDDERS
Pennsylvania Game Commission	17	0%	0%	7.7
Pennsylvania State Police	7	29%	29%	4.4
Dept. of Conservation & Natural Resources	6	17%	33%	3.5
Department of Transportation	7	14%	43%	4.9

Department of Human Services	12	25%	58%	2.6
Dept. of Military & Veterans Affairs	10	20%	60%	2.3
PA Fish and Boat Commission	3	67%	67%	2.0
Department of Environmental Protection	3	67%	67%	1.7
Department of Corrections	12	50%	75%	2.6

BEST PERFORMER PA Game Commission 17 contracts in sample. Zero uncontested outcomes. An average of 7.7 bidders — the clear benchmark for every other agency. 0% Single-bid rate 0% 1-2 bid rate 7.7 Avg bidders	NEEDS ATTENTION Department of Corrections 12 contracts in sample. Half returned only one bidder. Three-quarters had 1-2 bidders. 50% Single-bid rate 75% 1-2 bid rate 2.6 Avg bidders
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SECTION 4 — VENDOR PATTERN

Case Study: Vendor A

Vendor A appeared in more contracts in this sample than any other single vendor — and illustrates precisely how low competition harms value for money.

CASE STUDY — FOOD & BEVERAGE DISTRIBUTOR

Vendor A — 9 contracts in this random sample

7

contracts with competition

Vendor A submitted the **highest (most expensive) bid** every time — and lost to cheaper competitors on all 7 occasions. Vendor A' price was on average **66% above the winning bid**.

2

contracts uncontested

Vendor A was the **only bidder** — and won by default, with no market check on its pricing

When the market was given a chance to work, it consistently rejected Vendor A' pricing. When outreach failed to attract competitors, Pennsylvania paid that price.

SECTION 5 — RECOMMENDATIONS

Actions to Close the Competition Gap

SHORT-TERM — THIS PROJECT

EXPAND OUTREACH TO SMALL AND DIVERSE BUSINESSES

The immediate focus of this project is improving vendor engagement for small and diverse businesses. This is not just an equity argument: it is a procurement performance argument. [The number of small businesses receiving federal contracts has fallen 40% since 2010. Strong supplier diversity programs generate 133% greater procurement ROI than those without \(The Hackett Group\)](#) — and when dollars flow to local small businesses, they recirculate in the local economy at a 1.9–2.6x multiplier. Every business that never learns about a solicitation or never becomes fully registered is a missed bid, and missed bids are what produce single-bid and low bid contracts. This data shows that, in Pennsylvania, adding even one competitor reduces prices by 16%. Expanding the vendor pool in this way directly addresses the root cause of single-bid contracts — and the evidence is clear that it pays for itself.

LONG-TERM ACTIONS

1

MEASURE IT

Set a formal KPI: percentage of contracts receiving 3 or more bids, reported annually by agency and published publicly. You cannot manage what you do not measure.

2

IMPROVE IT

Invest in proactive vendor outreach, particularly for agencies with high single-bid rates. Study and replicate the Game Commission's approach — it works at scale, within the same rules.

3

AUDIT IT

Conduct an independent review of procurement outreach at the Department of Corrections, PA Fish & Boat Commission, and Department of Environmental Protection — where single-bid rates reached 50–67% in this sample. The goal is not to identify underperformance, but to understand the barriers each agency faces:

Are vendor pools genuinely thin in their procurement categories?

Are outreach methods reaching the right suppliers?

Are registration barriers preventing businesses from participating?

4

STANDARDIZE IT

PA eMarketplace currently accepts bid tabulations in a wide array of formats — PDF, XLS, XLSX — with no standardized internal structure, column naming, or layout. This makes systematic monitoring of competition levels slow, unreliable, and difficult to scale. Introduce consistent digital submission guidelines: a mandated template with defined fields for bidder name, bid amount, and award status. This would enable automated competition reporting, reduce the risk of data errors, and make monitoring routine rather than resource-intensive.

This isn't hypothetical. An automated extraction approach was piloted across all 825 contracts in this study and returned a **45% error rate on validation** — a direct consequence of inconsistent formatting across agencies. The manual review in this report exists because automation wasn't reliable enough to trust.

METHODOLOGY & DATA NOTES

All figures are from a **stratified random sample of 100 contracts** manually reviewed from 825 competitive solicitations on PA eMarketplace (FY 2024–25). Sample was stratified proportionally by agency and file type. Each bid tabulation document was opened and reviewed directly — bidder counts and amounts recorded by hand. Margin of error: ±10% at 95% confidence^[1]. Academic reference: Coviello & Marinello (2014); Decarolis (2014).

An automated extraction approach was initially piloted across all 825 contracts but was found to have a **45% error rate** on validation. No automated data is used in this report. Contracts with price spread exceeding 500% of the highest bid are excluded from ratio analysis.

1 — DATA SOURCE

All data comes from **PA eMarketplace** (emarketplace.state.pa.us), the Commonwealth of Pennsylvania's official public procurement platform. The study covers **FY 2024–2025 competitive solicitations**. In total, 825 competitive solicitations were identified — contracts that went through a formal open-tender process meaning any qualified vendor could submit a bid.

Important distinction: this study examines only competitively tendered contracts. It does not include sole-source awards, emergency contracts, or contracts awarded by other mechanisms. A low number of bidders on a competitive contract therefore reflects a market engagement failure, not a deliberate procurement decision to limit competition.

2 — SAMPLE SELECTION & MARGIN OF ERROR [1]

A **stratified random sample** of 100 contracts was drawn, proportionally allocated by agency and by file type (PDF, XLS, XLSX). This ensures the sample reflects the actual distribution of contracts across the procurement system, rather than over-representing any single agency or format.

Margin of error formula:

$$MOE = 1.96 \times \sqrt{p \times (1-p) / n} \times \sqrt{(N-n) / (N-1)}$$

Where p = estimated proportion (0.25), n = 100, N = 825

Result: ±9.8% ≈ **±10% at 95% confidence**

3 — HOW DATA WAS COLLECTED

For each sampled contract, the researcher navigated to the bid tabulation page on PA eMarketplace, opened the attached document, and recorded: the number of bidders; the lowest bid amount (winning price); and the highest bid amount (most expensive qualifying submission). Contracts where no dollar amounts were available were excluded from price calculations but retained in bidder count analysis where possible.

Known limitation: Bid tabulation documents vary widely in format across agencies. Where ambiguity existed about total bids vs. line-item prices, the contract was excluded from price analysis rather than risk a misrecording.

4 — COMPETITION METRICS

Contracts are grouped by number of bidders: 1, 2, 3–4, and 5+. The **single-bid rate** is the proportion of contracts receiving exactly one bid — used as the primary competition health indicator. The **1–2 bid rate** includes contracts with one or two bids: with only one alternative price available, competitive pressure remains very limited even with two submissions.

5 — PRICE SAVINGS CALCULATIONS [2]

The price savings figures show how much lower the winning price is, on average, compared to the highest price submitted — grouped by competition level. For each group: calculate average highest bid and average lowest bid, then saving as % of highest = (avg highest – avg lowest) ÷ avg highest × 100.

WORKED EXAMPLE — 2-BIDDER CONTRACTS (THE 16% FIGURE)

17 contracts with exactly 2 bidders (after outlier exclusion)

Average highest bid: $\$1,100,648 \div 17 = \$64,744$

Average lowest bid: $\$928,009 \div 17 = \$54,594$

Saving as % of highest: $(\$64,744 - \$54,594) \div \$64,744 \times 100 = 15.7\% \approx 16\%$

Interpretation note: The 3–4 bidder group shows a large average saving in dollar terms because it includes several large construction contracts which pull the average up. The percentage figures are more reliable cross-group comparisons than the raw dollar averages.

5B — WHY 5+ BIDDERS AS THE COMPETITIVE BENCHMARK

This study uses 5+ bidders as the “fully competitive” threshold for three reinforcing reasons:

Diminishing returns above ~5 bidders. The largest price reductions occur moving from 1 to 2 bidders, with meaningful further gains up to 3–4. Beyond 5 bidders, marginal improvements plateau — making 5+ a meaningful ceiling, not an arbitrary cut-off.

An achievable policy target. A benchmark of 10 or 15 bidders would be aspirational but unrealistic. Five or more bidders is within reach through improved vendor outreach and a simpler registration process — a demanding but credible bar for reform.

Data-driven grouping. Contract volume in this sample thins significantly above 5 bidders, making finer breakdowns statistically unreliable. The 5+ group (n=26) is the largest robust high-competition tier available in the data.

Interpretation note: The 5+ benchmark captures the pricing observed in the best-performing contracts in this sample. It does not represent an absolute ceiling — some markets with more specialised suppliers will not reach 5 bidders — but it reflects what is achievable under normal procurement conditions in this dataset.

6 — ANNUAL SAVINGS EXTRAPOLATION (~\$34M) [4]

Rather than comparing average prices across bidder groups — which conflates competition effects with the fact that larger contracts naturally attract more bidders — this calculation applies the **additional savings rate** each group would gain by reaching 5+ bidder competition levels.

The 100 contracts in this sample cost Pennsylvania \$16M in total. If every one of those contracts had attracted 5 or more bidders, Pennsylvania would have paid around \$4M less. Scaled to the full portfolio of 825 contracts, that gap becomes approximately \$34M per year.

Scaling formula:

Estimated annual saving = Sample saving × (Total population ÷ Sample size)

= $\$4,080,000 \times (825 \div 100) = \sim\$33.7M \approx \$34M$

Important caveat: This figure assumes the unseen 725 contracts resemble the sampled 100 in their competition distribution and contract values. It also assumes all under-competitive contracts could achieve 5+ bidder pricing with better outreach, which will not be true for every category. Treat this as indicative of direction and order of magnitude, not a precise forecast.

7 — THE 7% RESEARCH FINDING [3]

Research across **3.5 million government contracts in Europe** found that single-bid contracts are both a governance risk and **over 7% more expensive** than contracts that attracted competitive responses. Source: [Open Contracting Partnership — Evidence on procurement transparency and competition](#)

HOW THIS IS APPLIED TO PA DATA

Single-bid contracts in sample: 24 contracts · Total spend: \$1,778,000

Estimated overpayment at 7% premium: $\$1,778,000 \times 0.07 = \sim\$124,000$

PA’s own data suggests the true premium is at least 16% — more than twice the research benchmark.

8 — AGENCY PERFORMANCE METRICS

Agency-level figures are calculated from the subset of sampled contracts attributed to each agency. Only agencies with 3 or more contracts in the sample are reported individually — below this threshold, single contracts can swing percentages dramatically. The single-bid rate for each agency is that agency’s single-bid contracts divided by its total sampled contracts.

Small sample caveat: Agencies represented by only 3 contracts (e.g., Fish and Boat Commission, Environmental Protection) should be interpreted cautiously. Figures are directionally informative but should not be treated as precise agency-level measurements without a larger dedicated sample.

9 — WHAT THIS STUDY CANNOT TELL US

1. **Why specific contracts attracted only one bidder.** Low response could reflect insufficient outreach, a genuinely limited supplier market, overly restrictive specifications, or other factors. This study identifies the problem; it does not diagnose the cause for individual contracts.
2. **Whether single-bid prices are unreasonable.** A single bidder may still submit a market-rate price. The concern is that without competition, there is no mechanism to confirm this.
3. **Whether all single-bid contracts could attract more competition.** Some categories genuinely have few qualified suppliers. Savings estimates are indicative, not guaranteed outcomes.
4. **Contract quality and delivery outcomes.** This study focuses solely on price competition and does not assess whether lower-bidding vendors delivered satisfactory outcomes.
5. **Trends over time.** This study covers one fiscal year (FY 2024–25). Whether competition levels have improved or worsened over time cannot be determined from this data alone.